

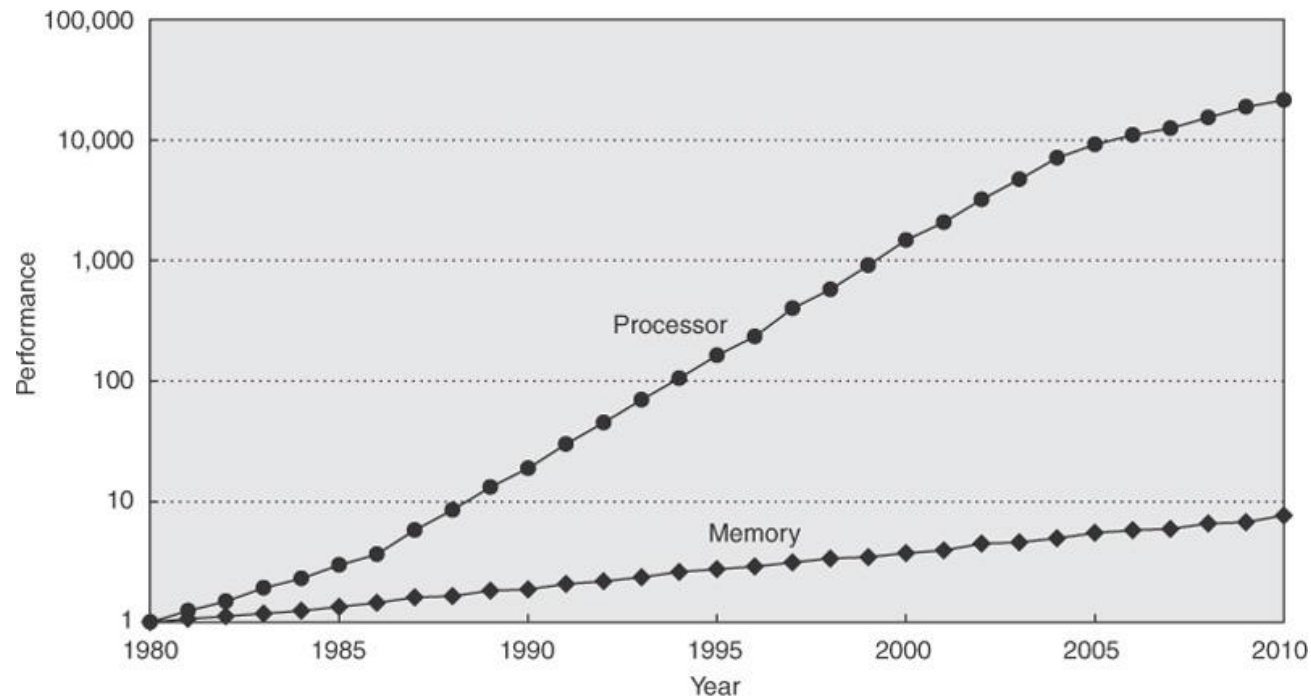
Advanced Cache System

Outline

- Non-blocking Cache
- Exclusive Cache
- Inclusive Cache
- Cache Coherence Protocol

Fast Processor v.s. Slow Memory

- Performance gap between processor and memory is increasing
 - Cache is essential to reduce processor stall cycles

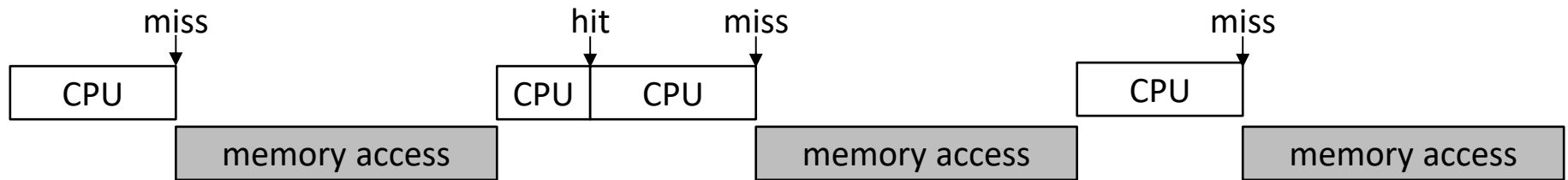


Improving Cache Performance

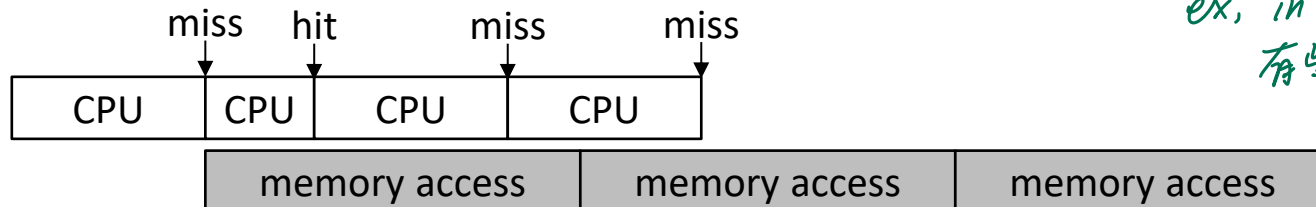
- Reduce miss count
 - Increase cache size
 - Increase cache associativity
 - Use multi-lateral cache (ex: victim cache)
- Reduce miss latency
 - Use multi-level cache hierarchy
 - Use **non-blocking cache**
 - Allow **multiple outstanding misses**

Blocking and Non-blocking Cache

- Blocking cache
 - CPU must be stalled on a cache miss



- Non-blocking cache
 - CPU may continue execution on outstanding miss



*ex, in an out-of-order execution,
有些被停下來, 有些繼續跑.*

Miss Status Holding Register (MSHR)

- Keep track of cache miss
- On a cache miss
 - MSHR miss
 - Allocate a MSHR entry
 - MSHR hit
 - Wait for data

Each MSHR entry 包含:

① 引起 miss 的 cache block address

② target information

valid	type	block offset	dest
-------	------	--------------	------

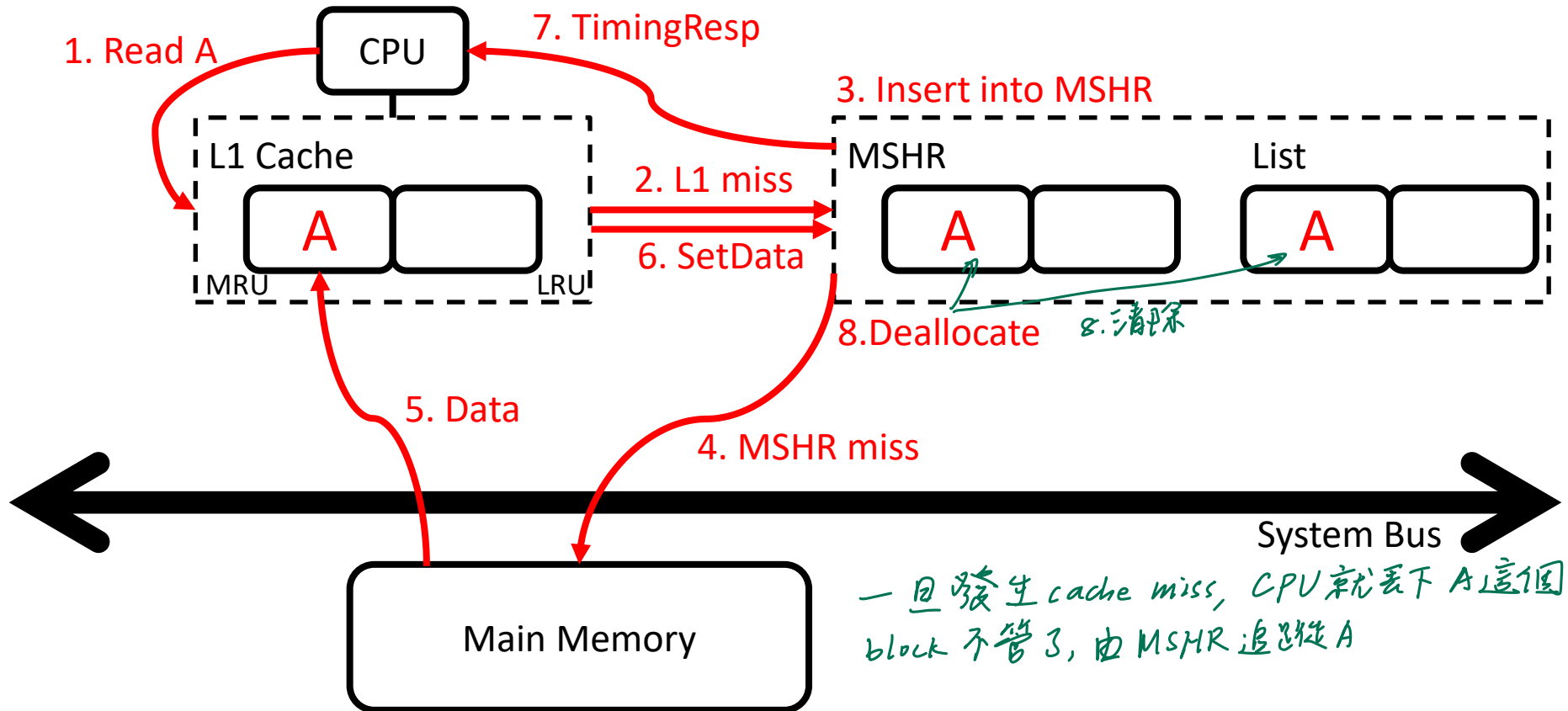
 (type: load/store)

③ A valid bit. 若所有 MSHR entry 的 valid bit 皆为 1, 再出现 miss, 只能 block.

若 requested data 比一个 cache line 小,
就需要 requested data 在该 cache line 的 offset

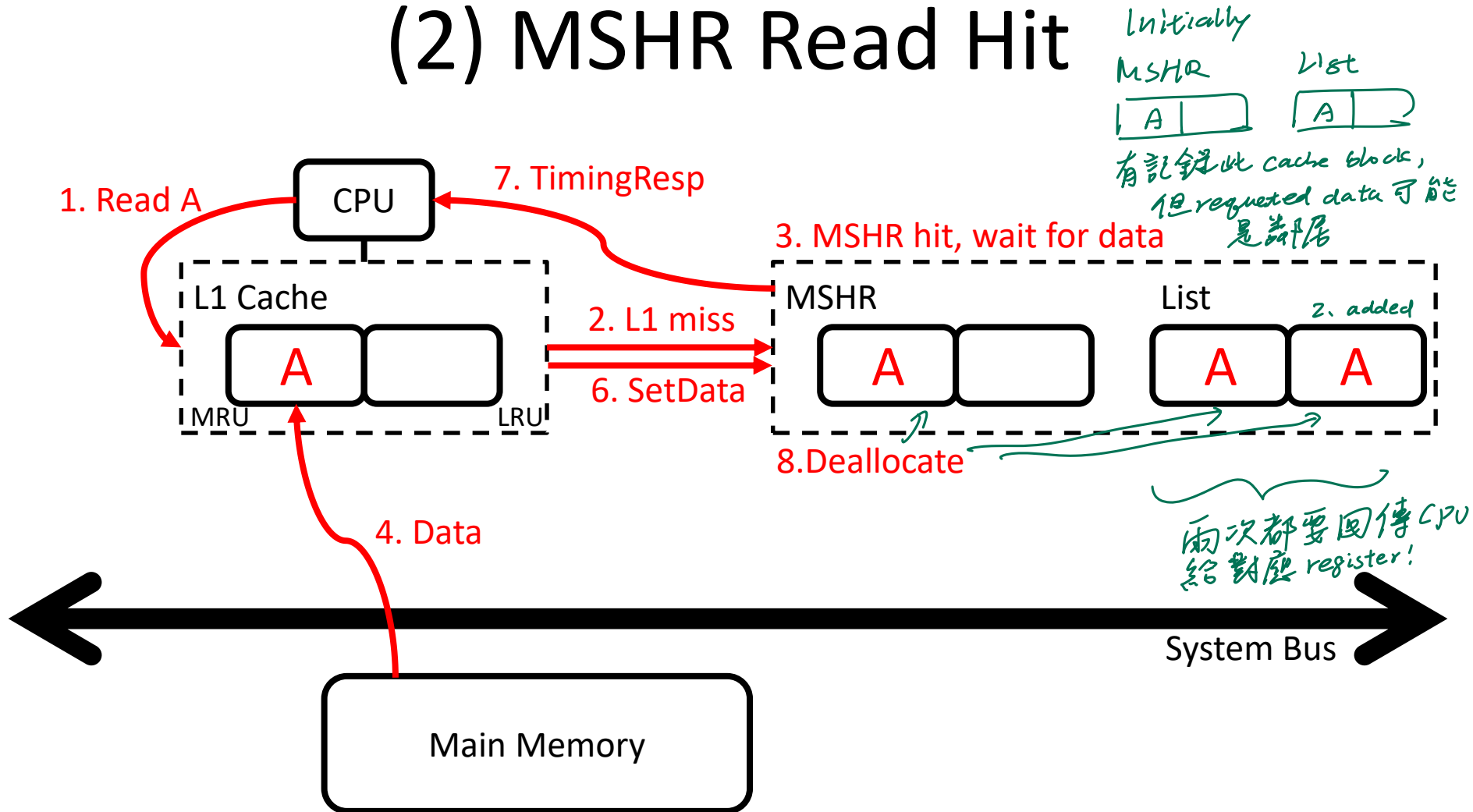
Miss Status Holding Registers

(1) MSHR Read Miss



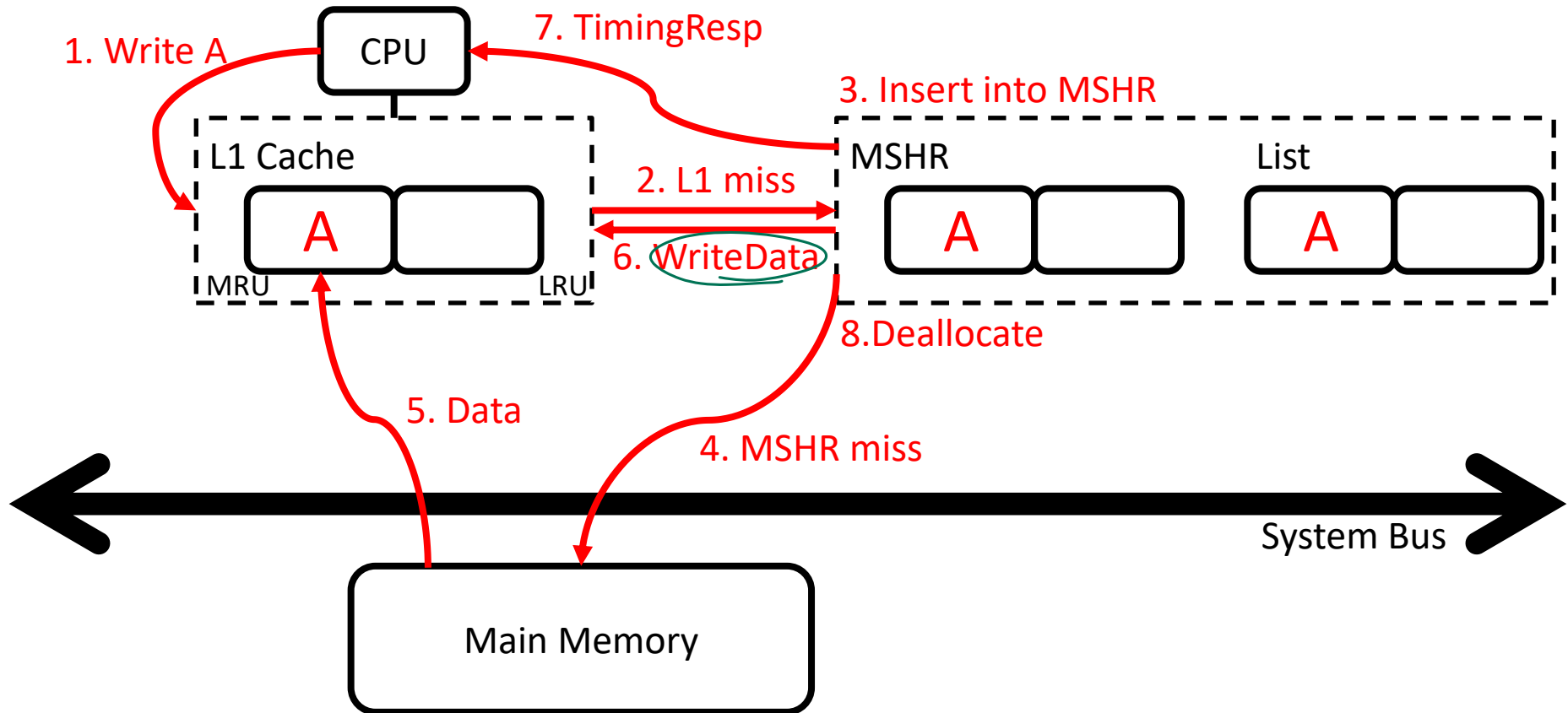
Miss Status Holding Registers

(2) MSHR Read Hit



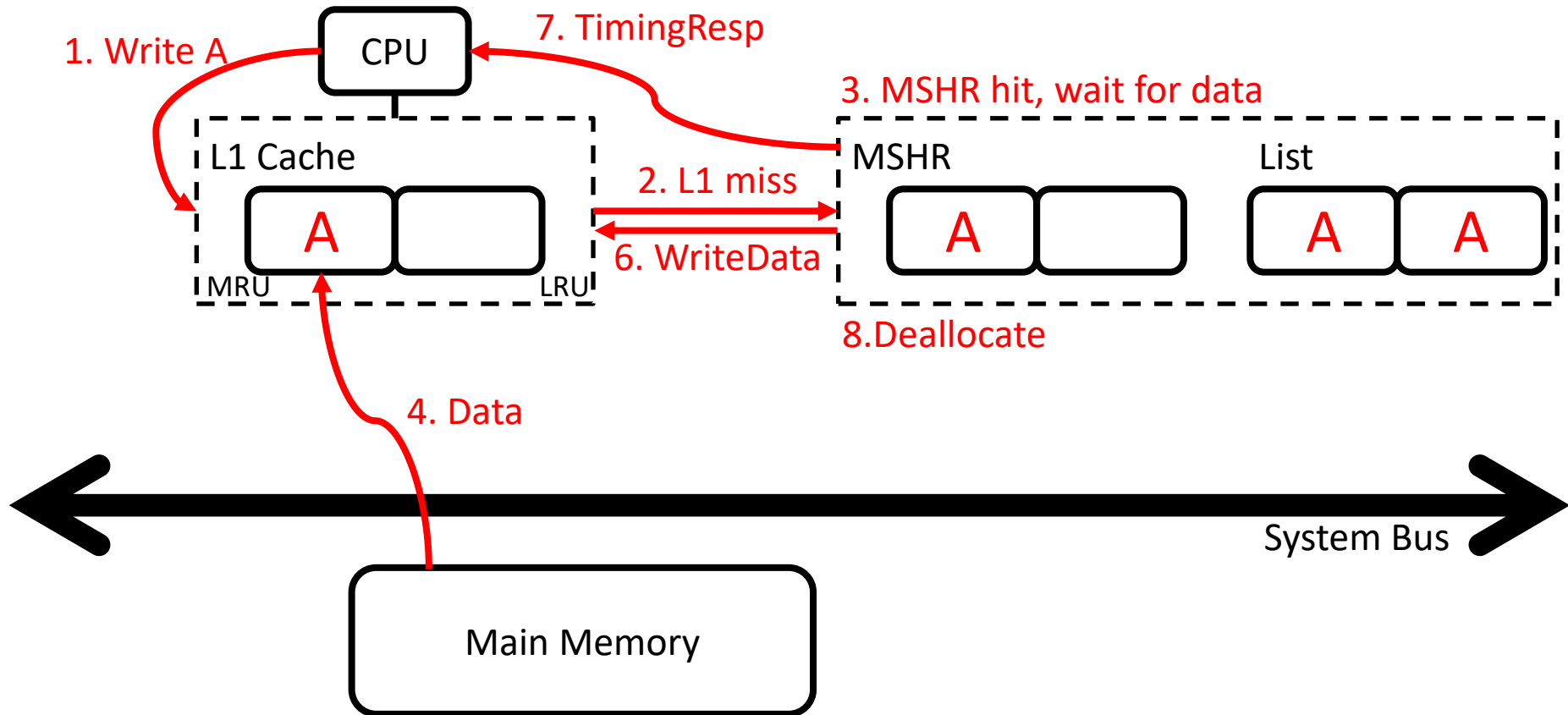
Miss Status Holding Registers

(3) MSHR Write Miss



Miss Status Holding Registers

(4) MSHR Write Hit



Outline

- Non-blocking Cache
- Exclusive Cache
- Inclusive Cache
- Cache Coherence Protocol

Strictly-exclusive Cache

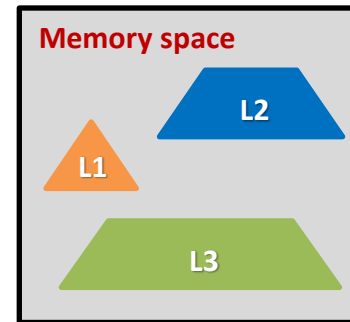
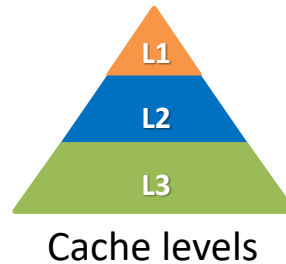
- Data is guaranteed to be in **at most one** level of the cache hierarchy

- Advantage

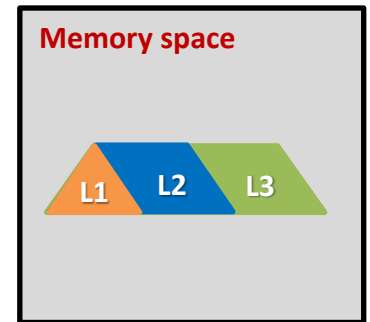
- Store more data

- Disadvantage

- Complicated cache coherence
 - Check all cache levels for coherency invalidation
 - Cache block size is fixed among all cache levels
 - Swap cache blocks between levels on a lower level cache hit



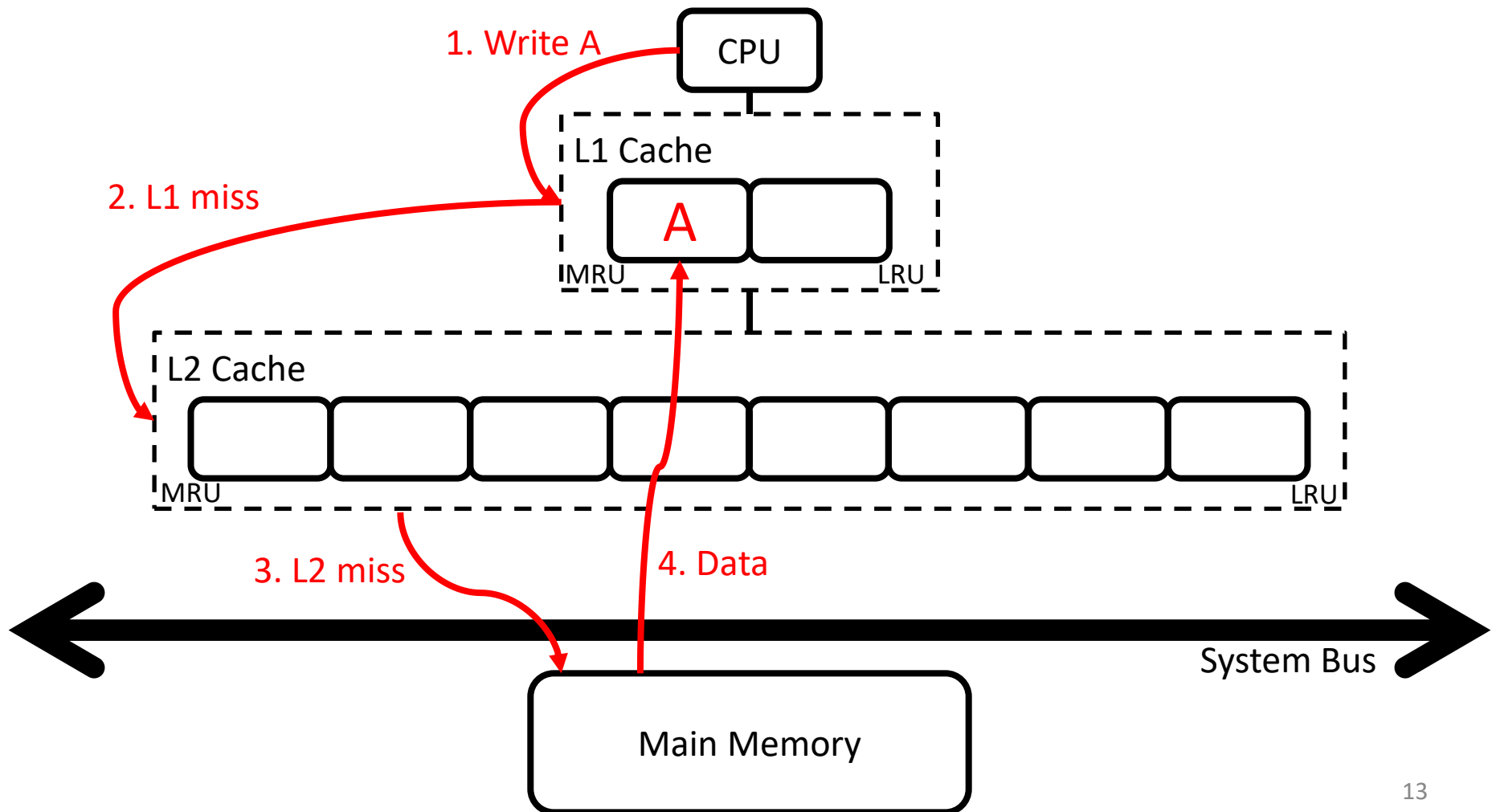
Exclusive cache



Inclusive cache

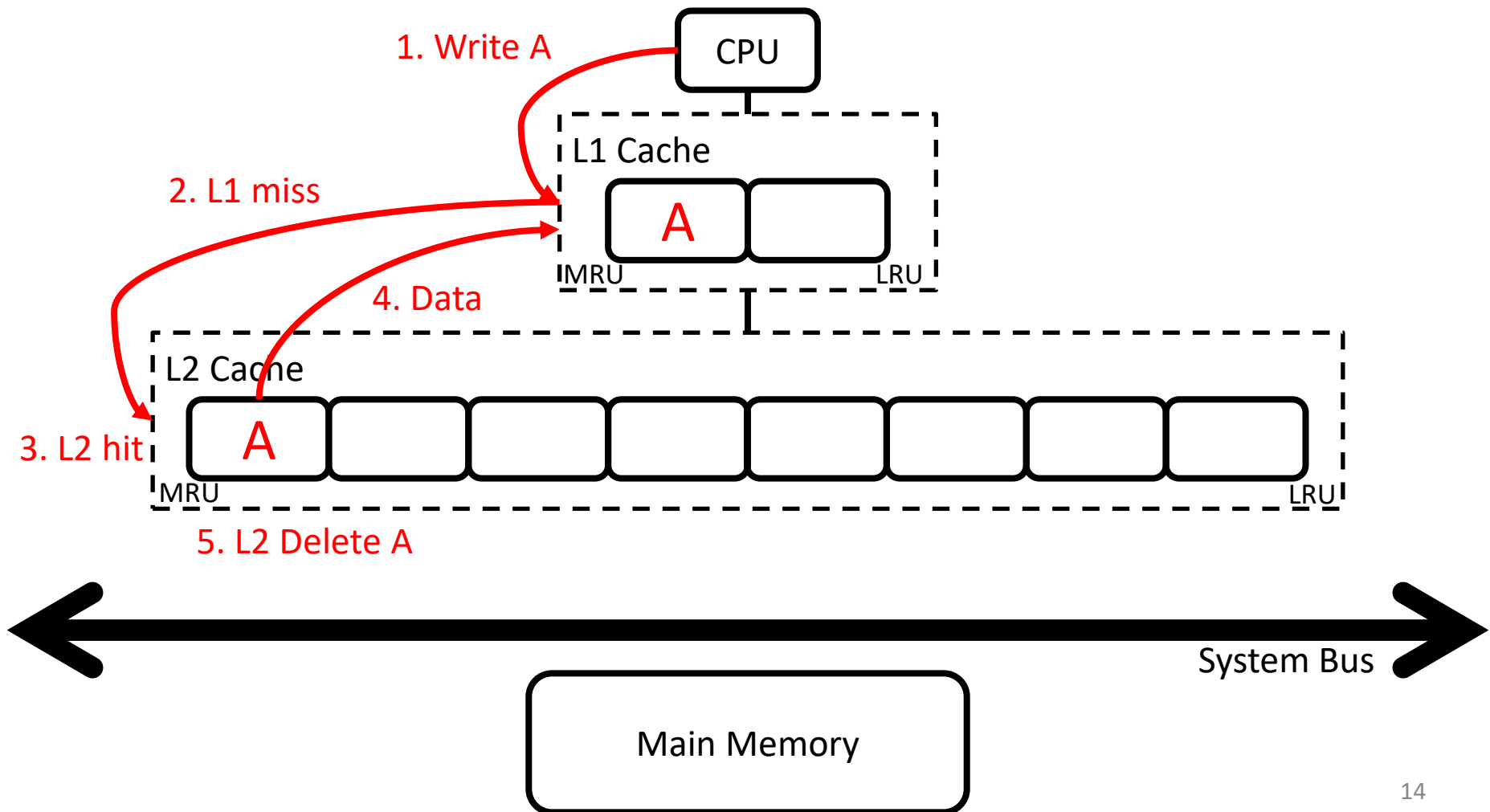
Strictly-exclusive Cache Behavior

(1) Data from Memory to L1



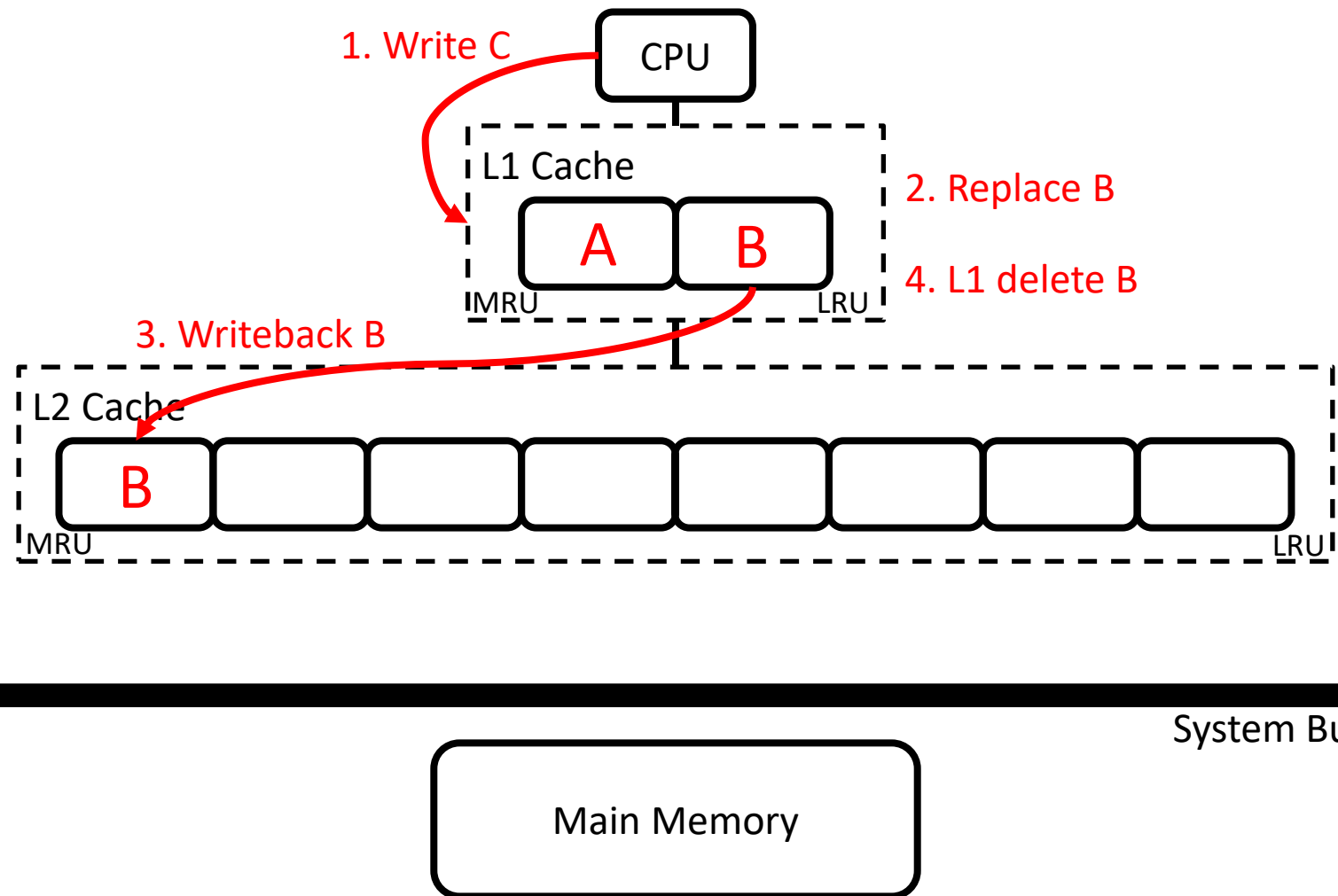
Strictly-exclusive Cache Behavior

(2) Data from L2 to L1



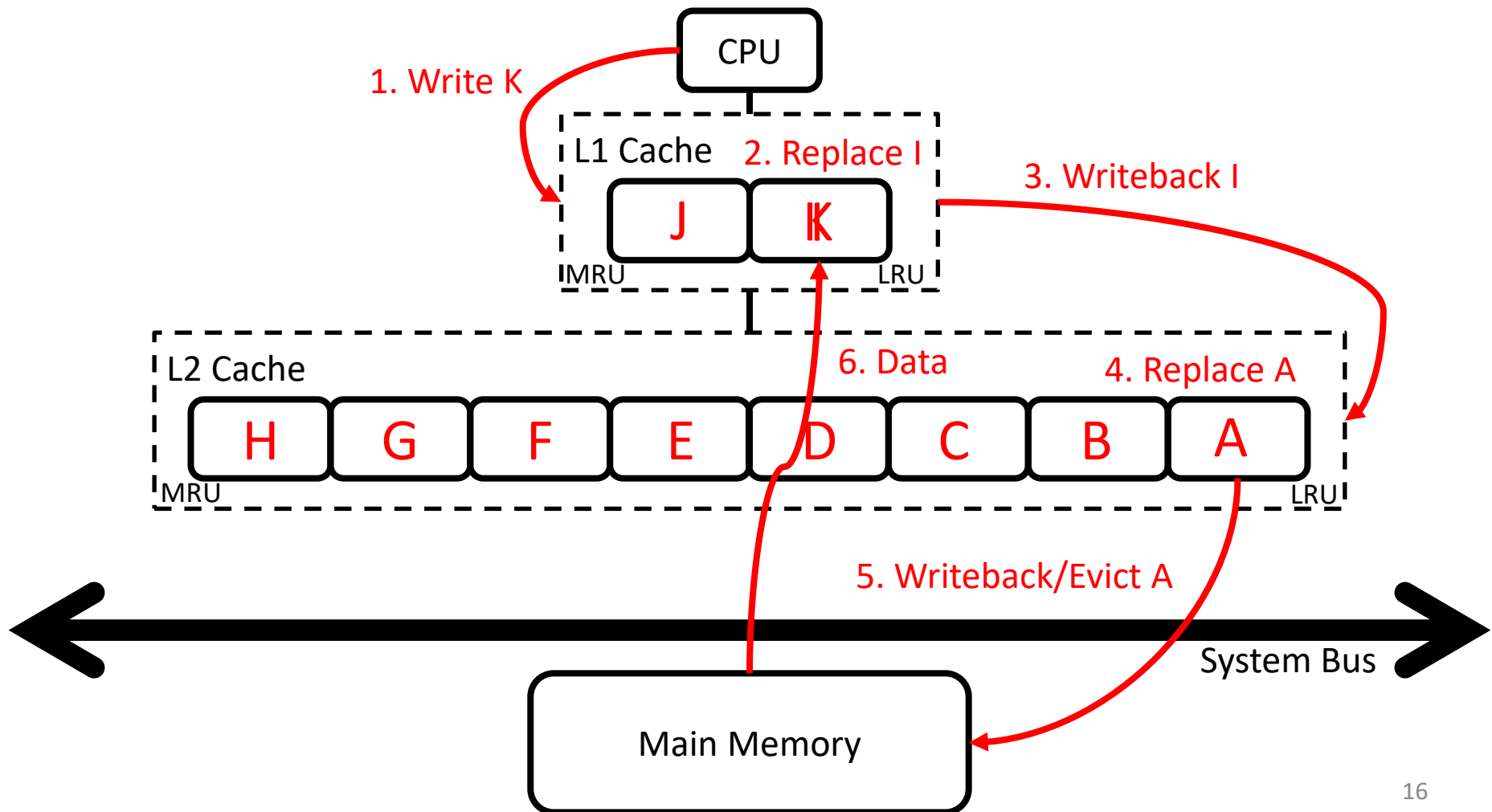
Strictly-exclusive Cache Behavior

(3) L1 Replacement



Strictly-exclusive Cache Behavior

(4) L2 Replacement

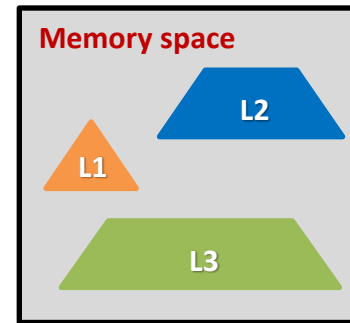
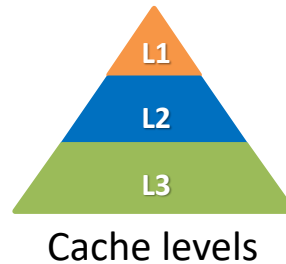


Outline

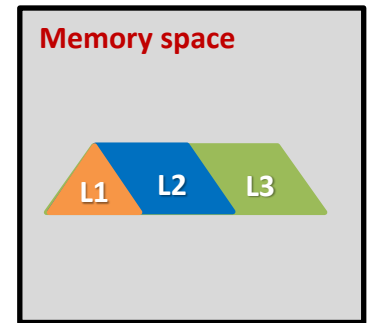
- Non-blocking Cache
- Exclusive Cache
- Inclusive Cache
- Cache Coherence Protocol

Strictly-inclusive Cache

- Data block in a higher level cache must be present somewhere in a lower level cache



Exclusive cache

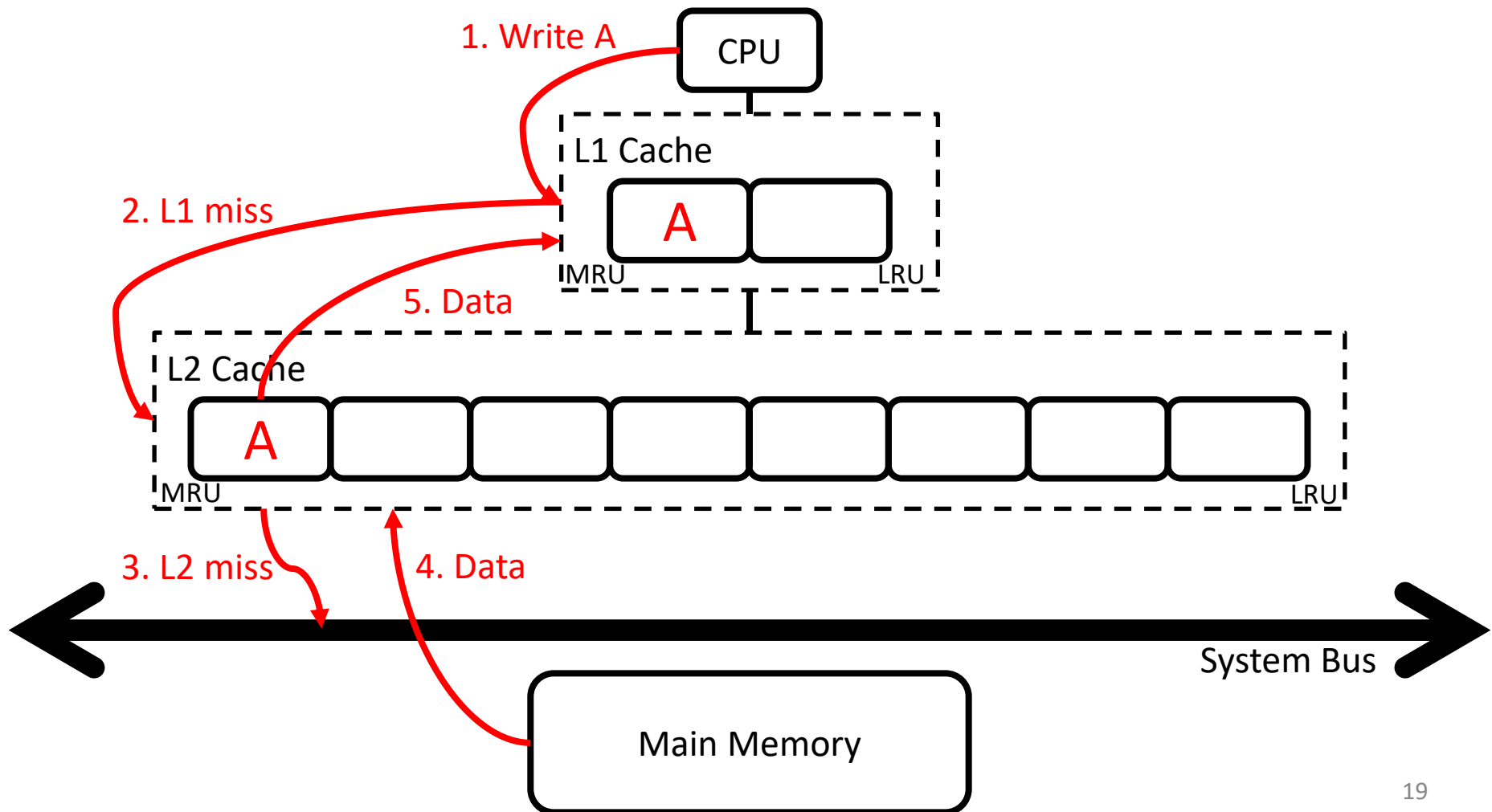


Inclusive cache

- Advantage
 - Simplified cache coherence
 - Check lower level cache for coherency invalidation
 - Variable cache block size among all cache levels
- Disadvantage
 - Evict corresponding L1 block on a L2 block eviction
 - Maintain inclusive property results in higher miss rate
 - Store less data

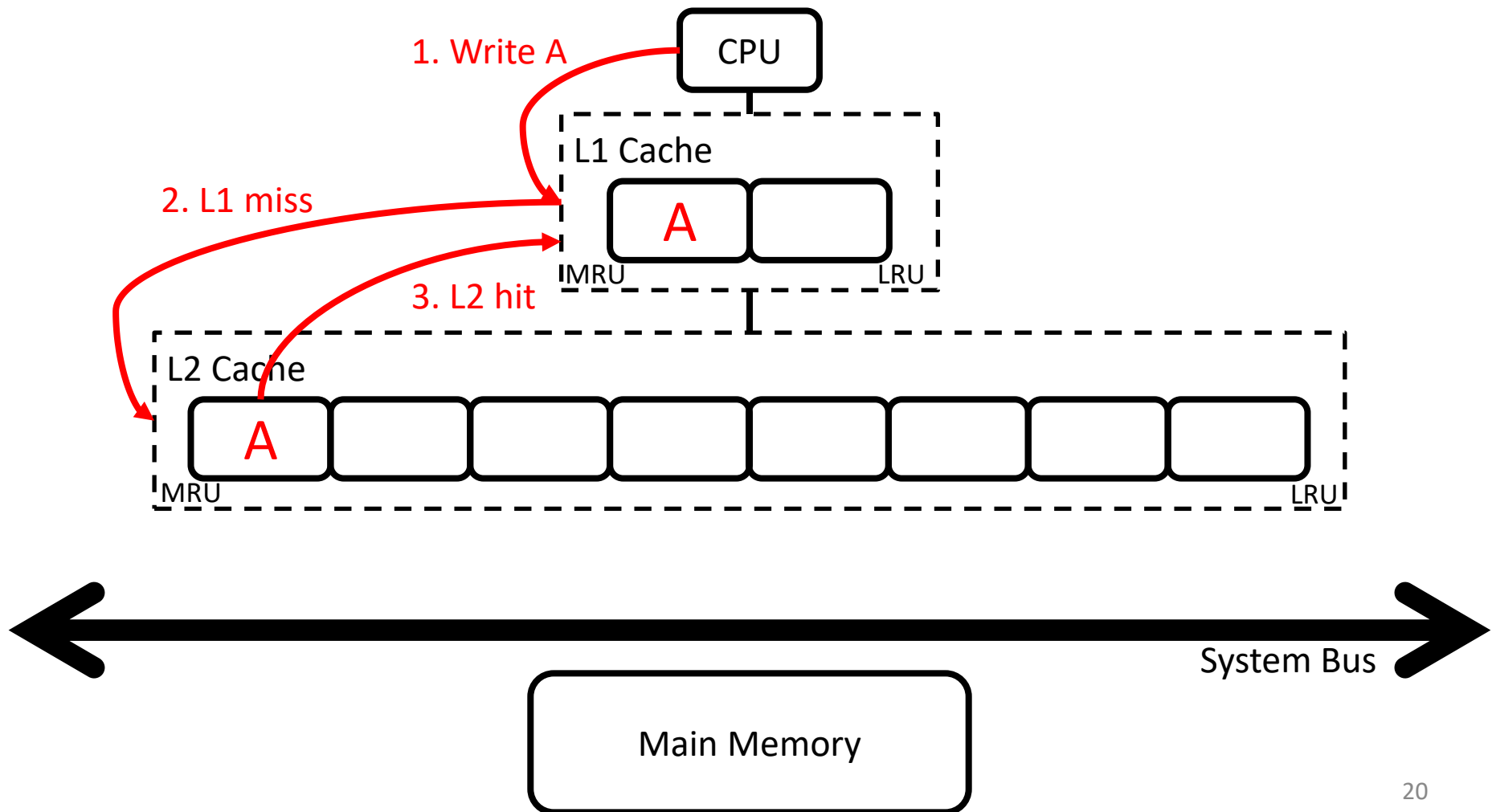
Strictly-inclusive Cache Behavior

(1) Data from Memory to L1



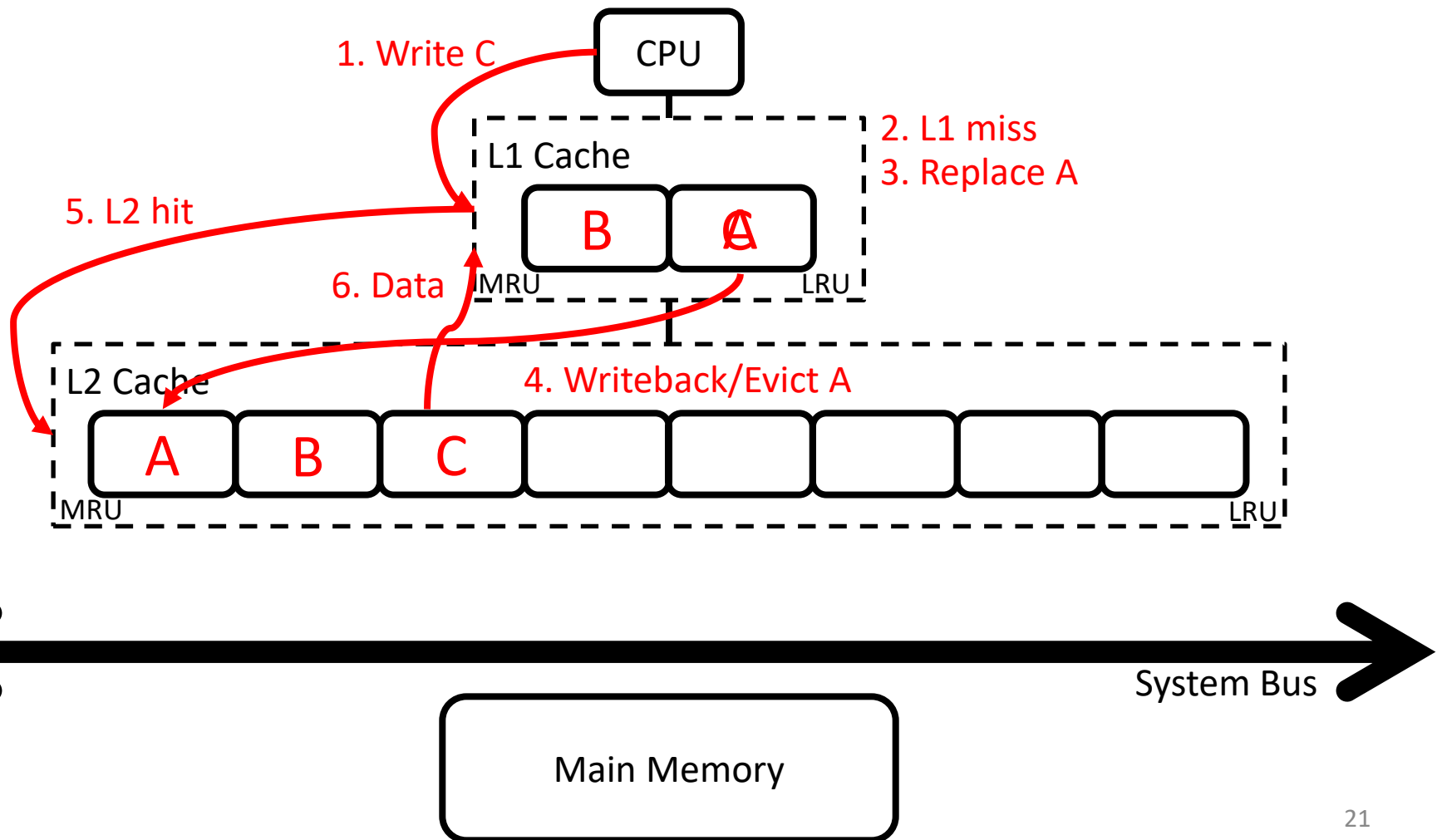
Strictly-inclusive Cache Behavior

(2) Data from L2 to L1



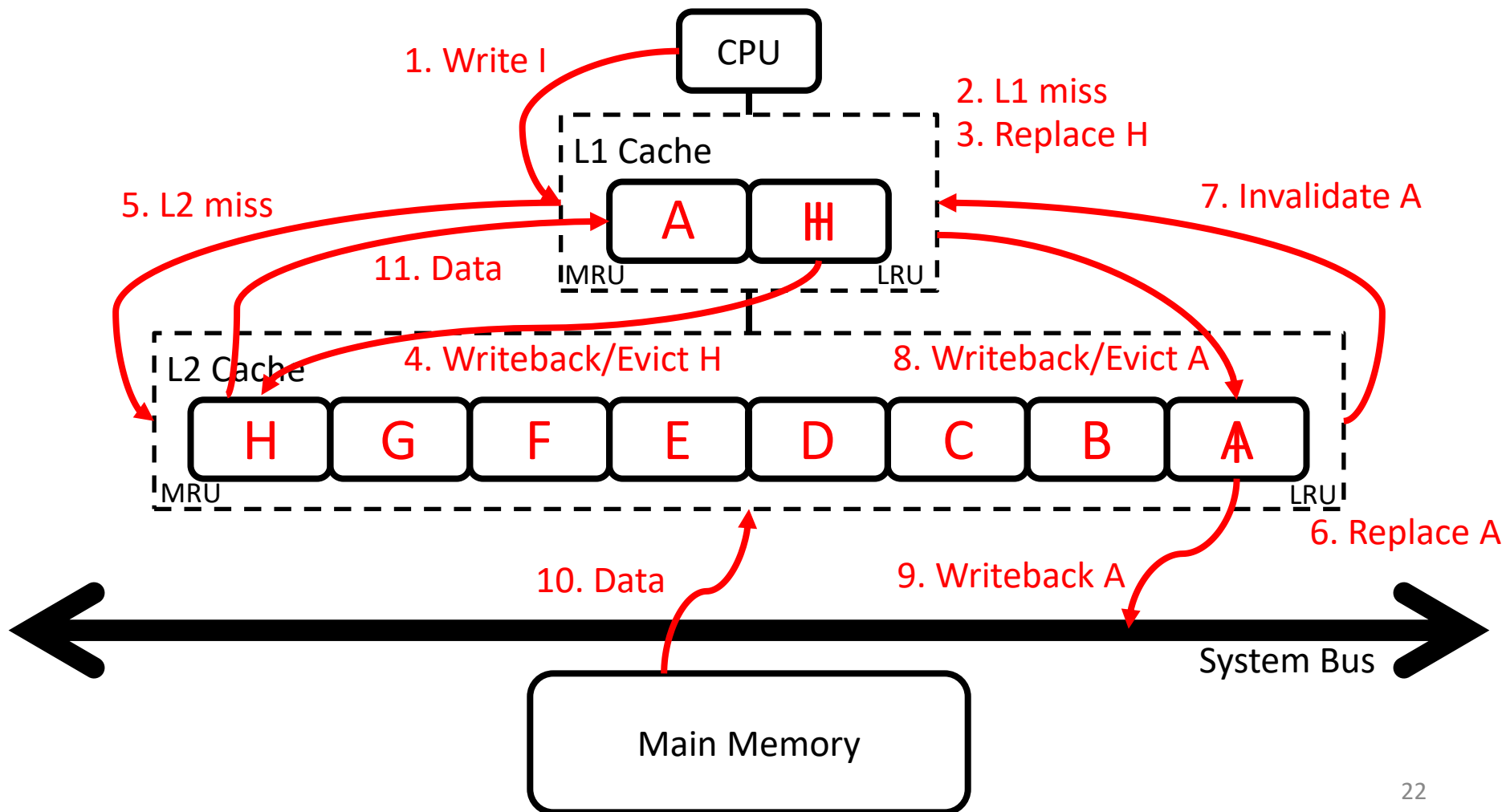
Strictly-inclusive Cache Behavior

(3) L1 Replacement



Strictly-inclusive Cache Behavior

(4) L2 Replacement



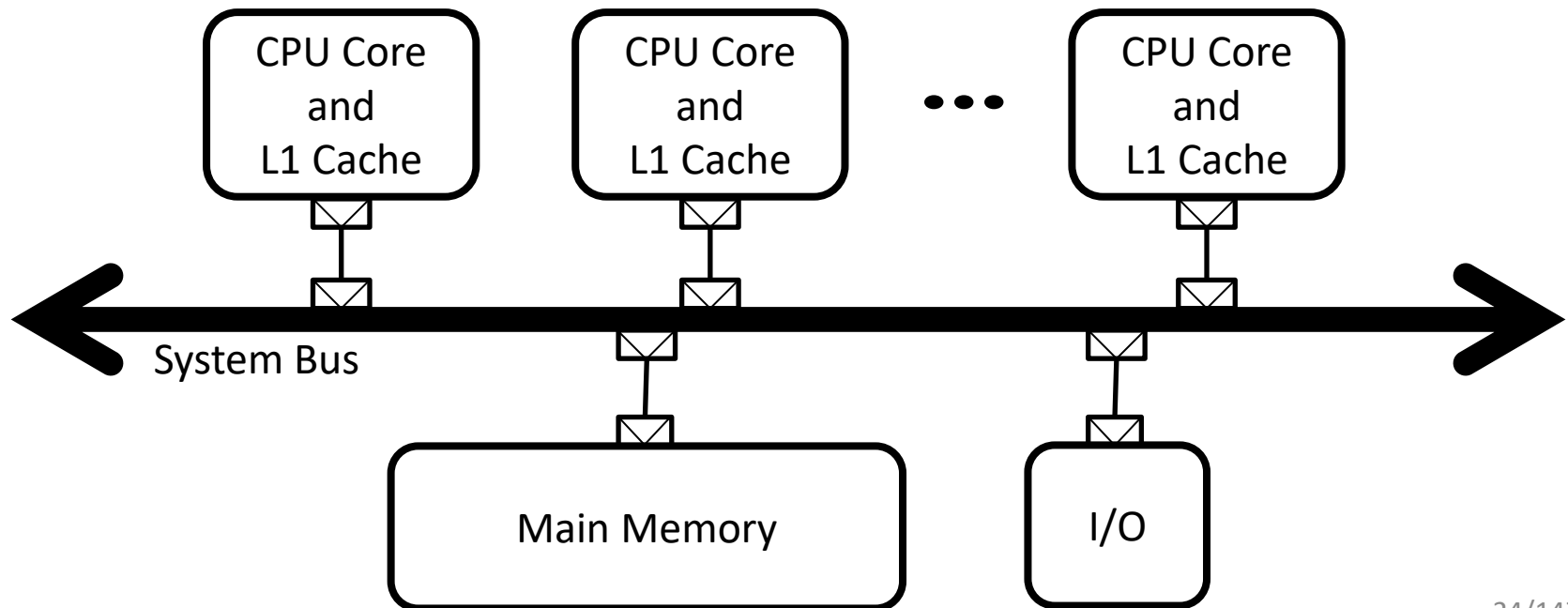
Outline

- Non-blocking Cache
- Exclusive Cache
- Inclusive Cache
- Cache Coherence Protocol

需要 cache coherence “多核” multi-core

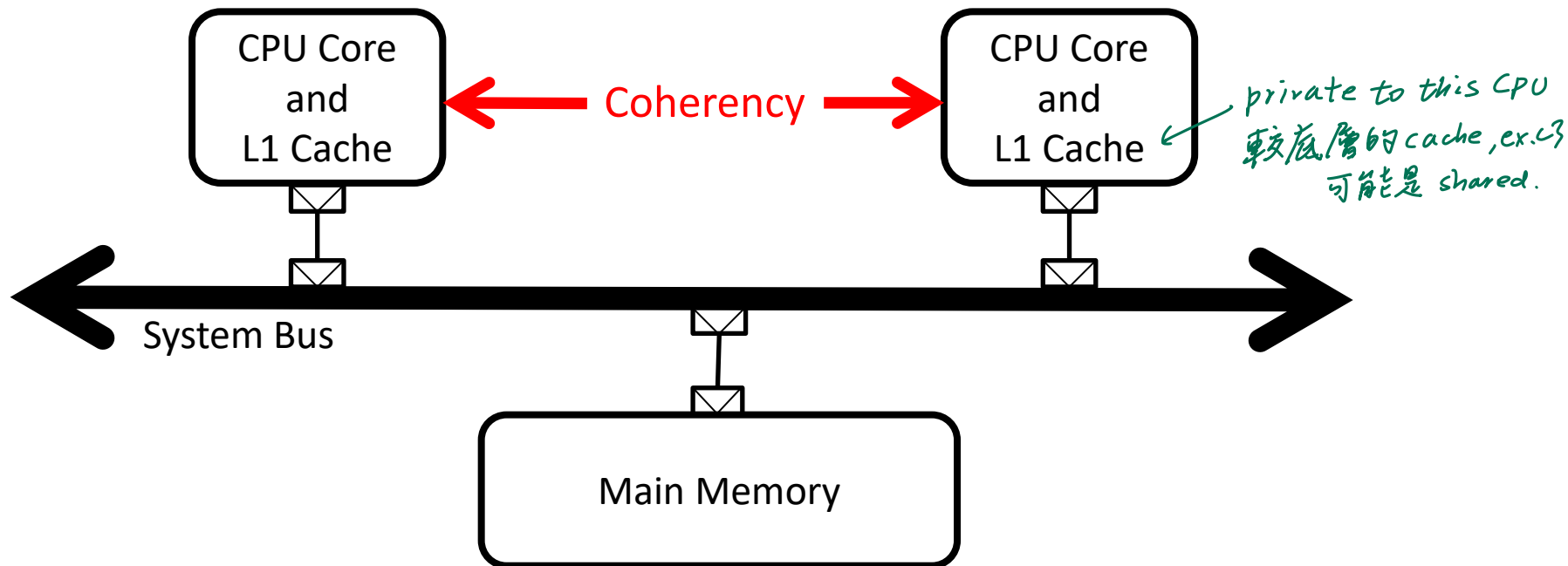
Multiprocessing

- The use of two or more central processing units (CPUs) within a single computing system
 - Sharing memory resource



Cache Coherence

- The consistency of shared data that stored in multiple local (private) caches



Cache Coherence Protocol

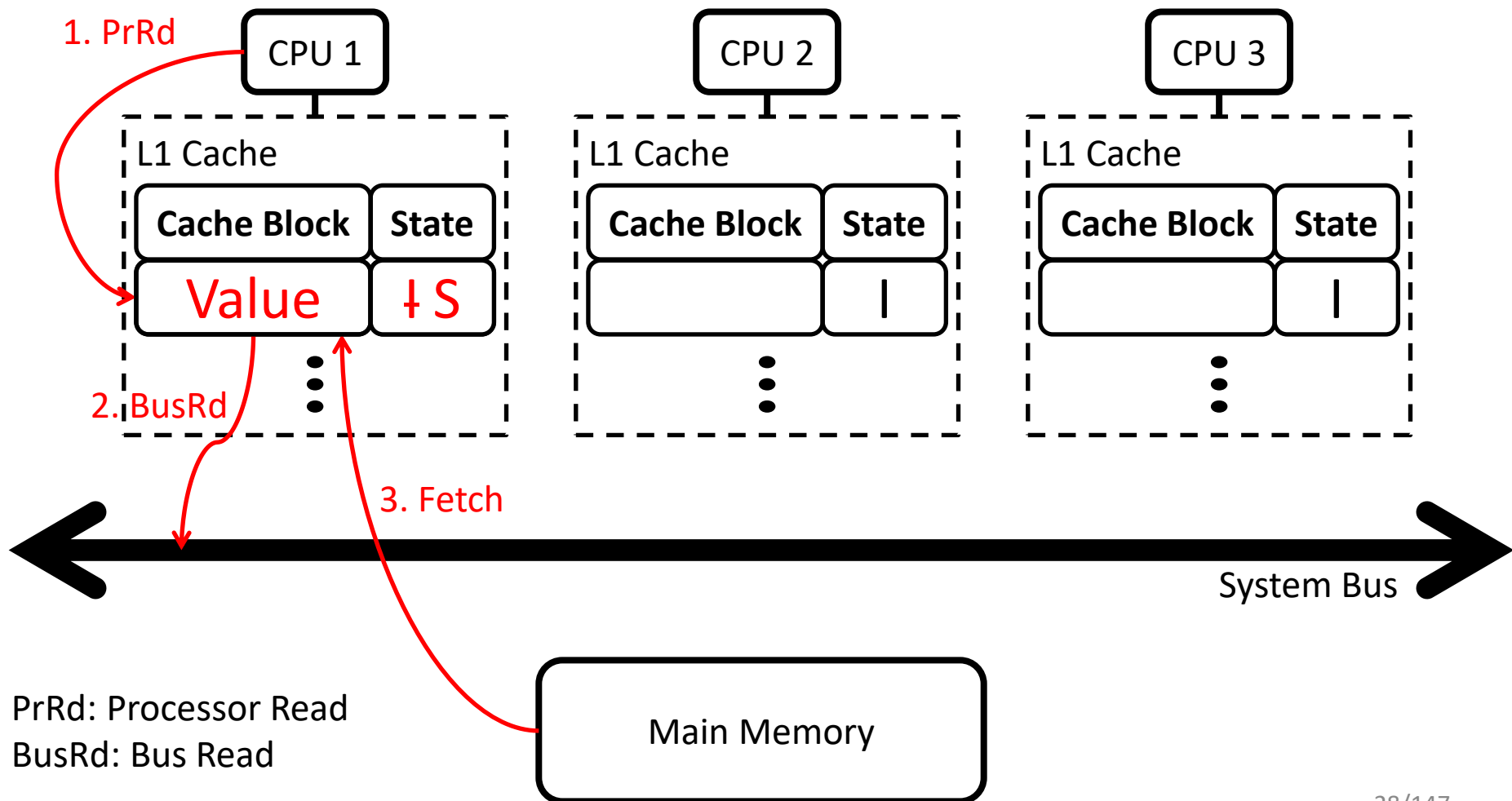
- Maintain the consistency among all private caches in a shared-memory system
 - MSI protocol
 - MESI protocol
 - MESIF protocol
 - MOESI protocol
 - ...

MSI Protocol

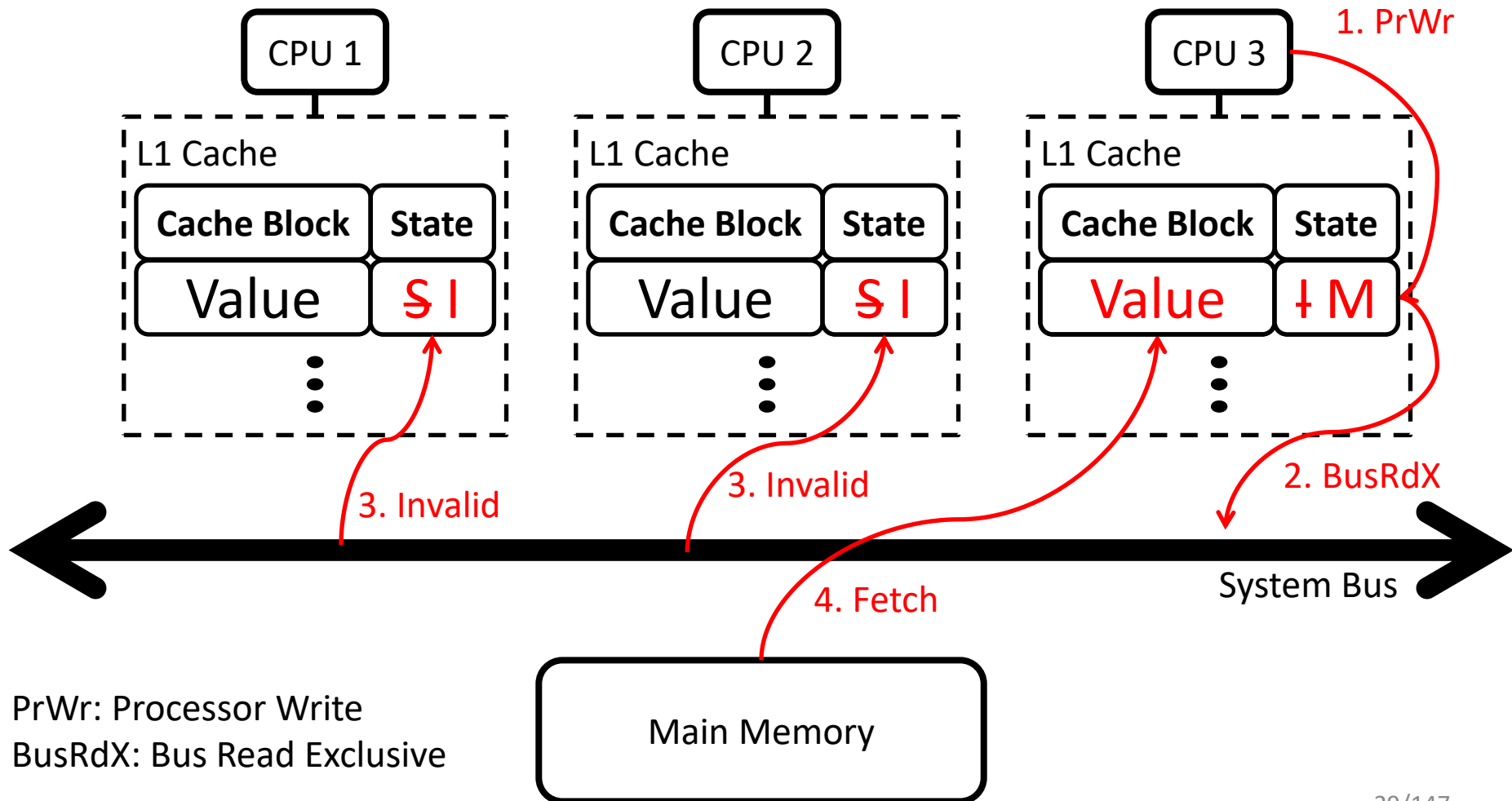
- Three main states
 - **M**odified: cache block has been modified (*dirty*)
 - **S**hared: cache block is unmodified and exists in at least one cache
 - **I**nvalid: cache block is invalid

state 記錄在每一個 cache block 中.

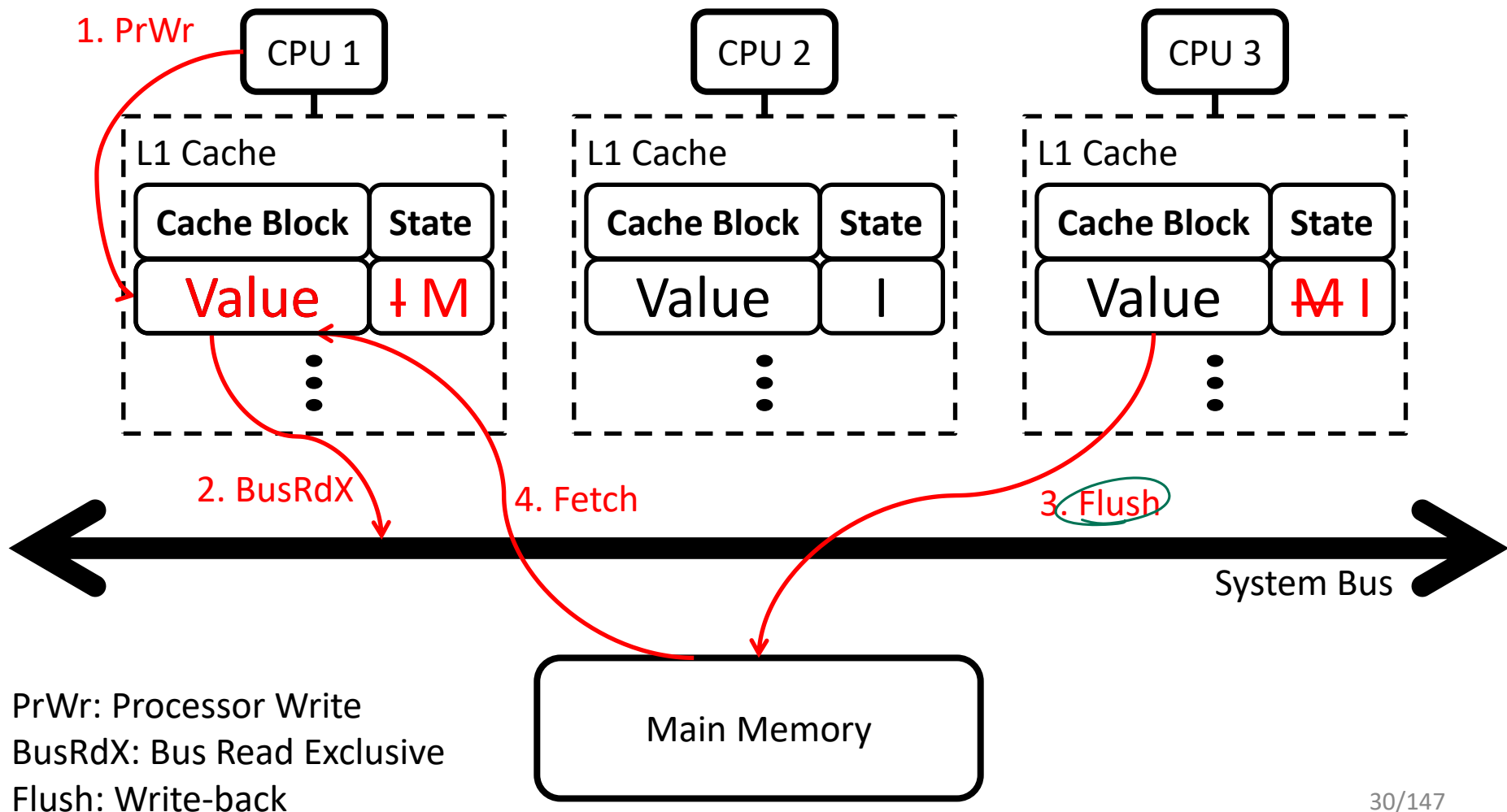
MSI Protocol Example 1/4



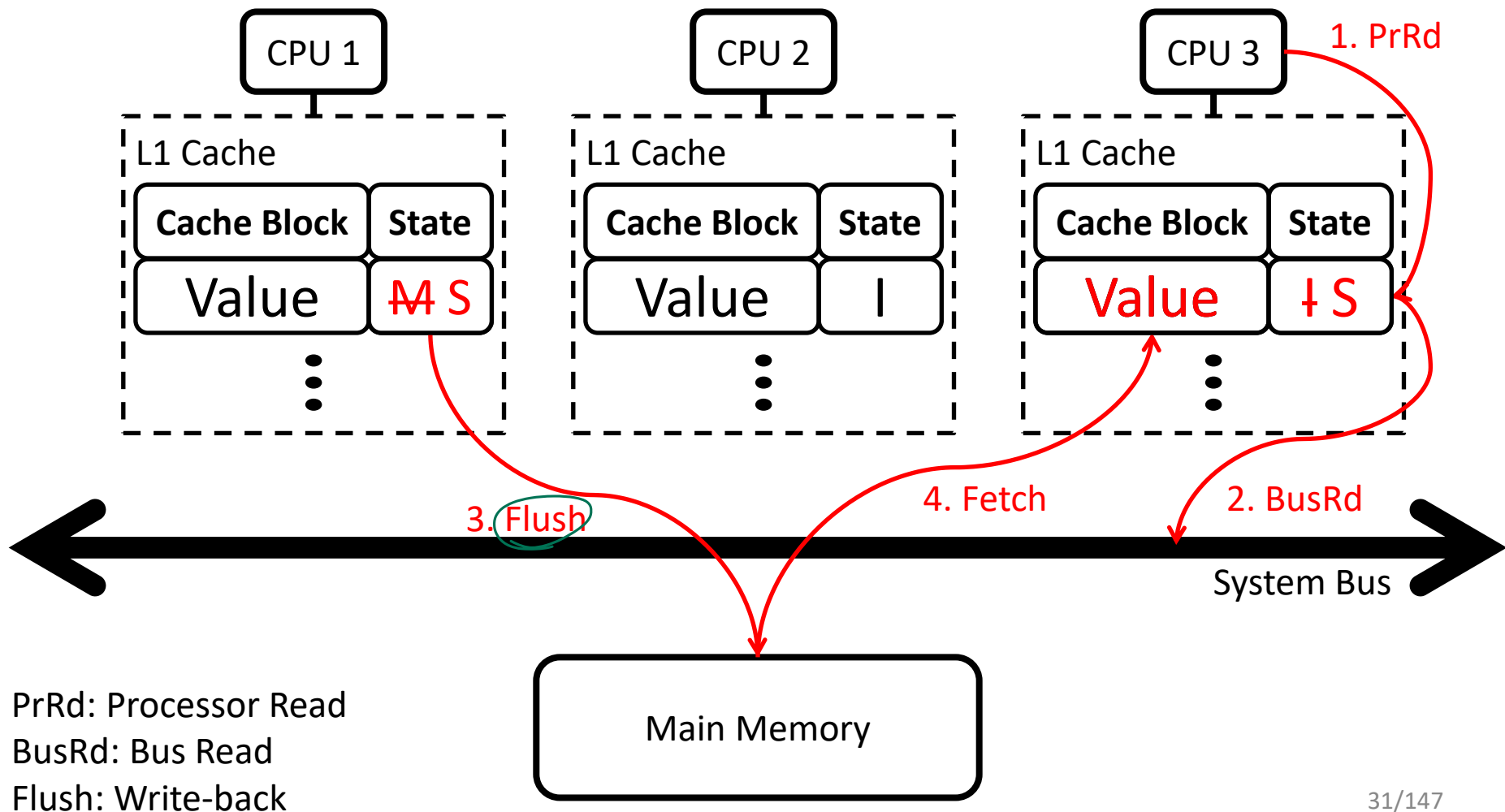
MSI Protocol Example 2/4



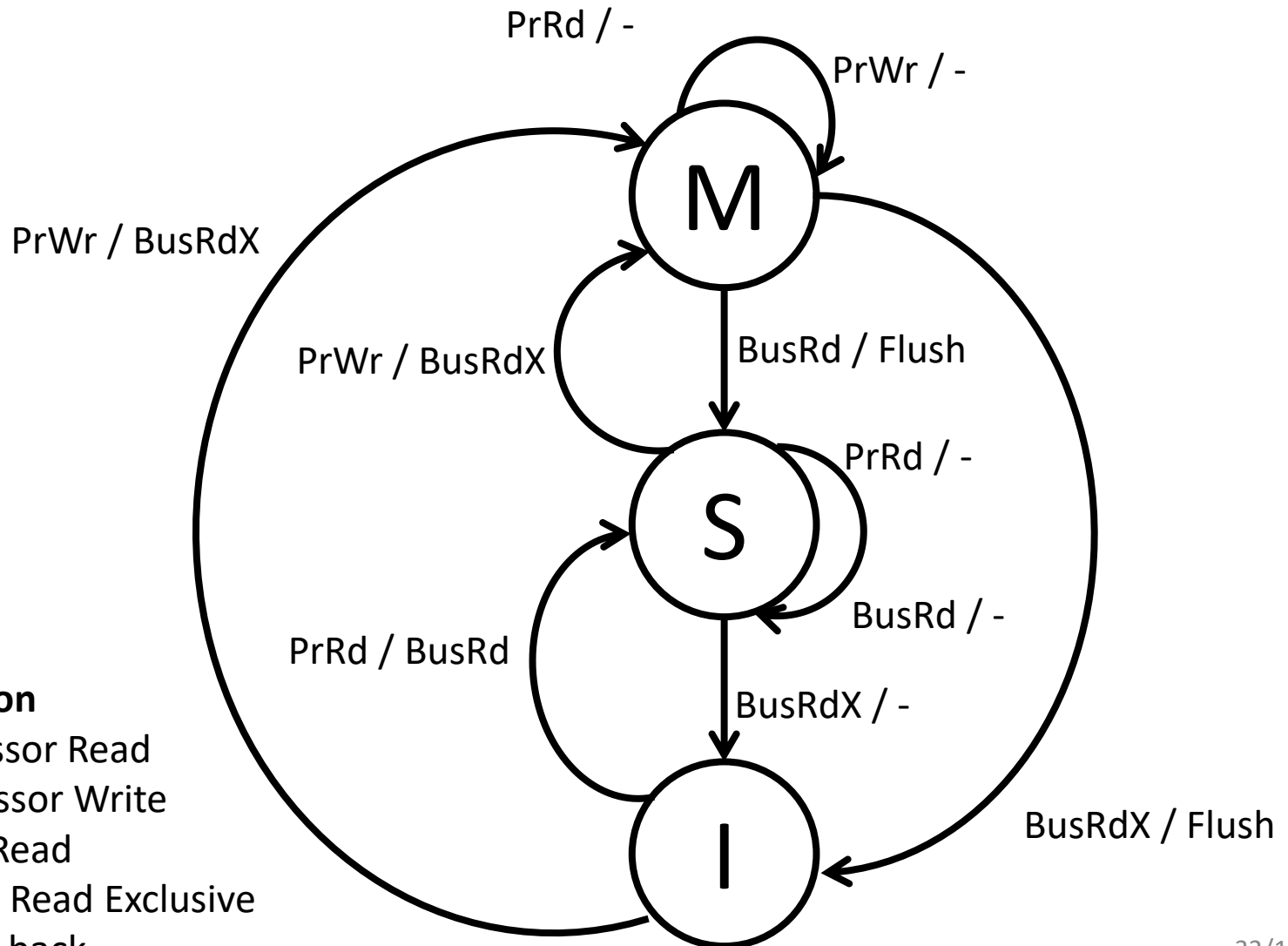
MSI Protocol Example 3/4



MSI Protocol Example 4/4



MSI State Diagram



Event / Action

PrRd: Processor Read

PrWr: Processor Write

BusRd: Bus Read

BusRdX: Bus Read Exclusive

Flush: Write-back

Cache Coherence Protocols

- Maintains the consistency between all the caches in a system of distributed shared memory
 - MSI protocol
 - MESI protocol
 - MESIF protocol (Intel-alike)
 - MOESI protocol (AMD-alike)
 - ...etc.

MESI Protocol

- Four stable states
 - **M**odified: cache block has been modified
 - **E**xclusive: cache block is unmodified and exists in only one cache
 - **S**hared: cache block is unmodified and exists in at least one cache
 - **I**nvalid: cache block is invalid or not present in cache

MOESI Protocol

- Five stable states
 - **M**odified: cache block has been modified
 - **O**wned: cache block is owned by one cache
 - **E**xclusive: cache block is unmodified and exists in only one cache
 - **S**hared: cache block is unmodified and exists in at least one cache
 - **I**nvalid: cache block is invalid